

Classify Images with TensorFlow Convolutional Neural Networks

experiment Lab **schedule** 1 hour 15 minutes **universal_currency_alt** 5 Credits

show_chart Intermediate

★★★★★ [Rate Lab](#)

info This lab may incorporate AI tools to support your learning.

GSP633



Overview

A convolution is a filter that passes over an image, processes it, and extracts features that show a commonality in the image. A convolutional neural network (CNN) is a class of deep neural network (DNN) most commonly applied to visual imagery.

In this lab, you start with an image classification model developed from computer vision tools and then use CNNs to improve it. You'll use data (items of clothing) from a common dataset called Fashion MNIST.

Objectives

In this lab, you will learn how to:

- Review the the starting model
- Add convolutions, gather the data, and define the model

- Compile and train the model
- Visualize the convolutions and pooling

Prerequisites

To maximize your learning, consider taking these labs before taking this one:

- Introduction to Computer Vision with TensorFlow
- Introduction to Convolutions with TensorFlow

Setup and requirements

Before you click the Start Lab button

Read these instructions. Labs are timed and you cannot pause them. The timer, which starts when you click **Start Lab**, shows how long Google Cloud resources will be made available to you.

This hands-on lab lets you do the lab activities yourself in a real cloud environment, not in a simulation or demo environment. It does so by giving you new, temporary credentials that you use to sign in and access Google Cloud for the duration of the lab.

To complete this lab, you need:

- Access to a standard internet browser (Chrome browser recommended).

Note: Use an Incognito or private browser window to run this lab. This prevents any conflicts between your personal account and the Student account, which may cause extra charges incurred to your personal account.

- Time to complete the lab---remember, once you start, you cannot pause a lab.

Note: If you already have your own personal Google Cloud account or project, do not use it for this lab to avoid extra charges to your account.

How to start your lab and sign in to the Google Cloud console

1. Click the **Start Lab** button. If you need to pay for the lab, a pop-up opens for you to select your payment method. On the left is the **Lab Details** panel with the following:

- The **Open Google Cloud console** button
- Time remaining
- The temporary credentials that you must use for this lab
- Other information, if needed, to step through this lab

2. Click **Open Google Cloud console** (or right-click and select **Open Link in Incognito Window** if you are running the Chrome browser).

The lab spins up resources, and then opens another tab that shows the **Sign in** page.

Tip: Arrange the tabs in separate windows, side-by-side.

Note: If you see the **Choose an account** dialog, click **Use Another Account**.

3. If necessary, copy the **Username** below and paste it into the **Sign in** dialog.

"Username"

content_c

You can also find the **Username** in the **Lab Details** panel.

4. Click **Next**.

5. Copy the **Password** below and paste it into the **Welcome** dialog.

"Password"

content_c

You can also find the **Password** in the **Lab Details** panel.

6. Click **Next**.

Important: You must use the credentials the lab provides you. Do not use your Google Cloud account credentials.

Note: Using your own Google Cloud account for this lab may incur extra charges.

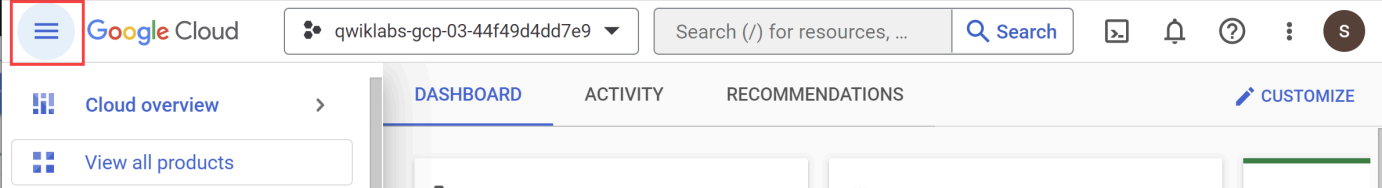
7. Click through the subsequent pages:

- Accept the terms and conditions.

- Do not add recovery options or two-factor authentication (because this is a temporary account).
- Do not sign up for free trials.

After a few moments, the Google Cloud console opens in this tab.

Note: To view a menu with a list of Google Cloud products and services, click the **Navigation menu** at the top-left.



Activate Cloud Shell

Cloud Shell is a virtual machine that is loaded with development tools. It offers a persistent 5GB home directory and runs on the Google Cloud. Cloud Shell provides command-line access to your Google Cloud resources.

1. Click **Activate Cloud Shell**  at the top of the Google Cloud console.

When you are connected, you are already authenticated, and the project is set to your **Project_ID**, `PROJECT_ID`. The output contains a line that declares the **Project_ID** for this session:

```
Your Cloud Platform project in this session is set to "PROJECT_ID"
```

`gcloud` is the command-line tool for Google Cloud. It comes pre-installed on Cloud Shell and supports tab-completion.

2. (Optional) You can list the active account name with this command:

```
gcloud auth list
```

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3. Click **Authorize**.

Output:

```
ACTIVE: *
ACCOUNT: "ACCOUNT"

To set the active account, run:
$ gcloud config set account `ACCOUNT`
```

4. (Optional) You can list the project ID with this command:

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```
gcloud config list project
```

Output:

```
[core]
project = "PROJECT_ID"
```

Note: For full documentation of `gcloud`, in Google Cloud, refer to the `gcloud` CLI overview guide.

Task 1. Open the notebook in Vertex AI Workbench

1. In the Google Cloud console, on the **Navigation menu** () , click **Vertex AI > Workbench**.
2. Find the `Workbench instance name` instance and click on the **Open JupyterLab** button.

The JupyterLab interface for your Workbench instance opens in a new browser tab.

Task 2. Run the lab notebook

1. In the left hand menu, modify `Notebook name` to include your region in cell 8. You can get it in the left panel of the lab instructions.

📁 / ... / learning-tensorflow / convolutional-neural-networks /

Name	Last Modified
📄 CLS_Vertex_AI_CNN_fmnnist.ipynb	2 minutes ago

2. Continue the lab in the notebook, and run each cell by clicking the **Run** () icon at the top of the screen. Alternatively, you can execute the code in a cell with **SHIFT + ENTER**.

Read the narrative and make sure you understand what's happening in each cell.

In order to view the status of training and deployment on Vertex AI, you can follow the instructions in the notebook containing illustrations.

Check your progress on notebook

Create a Cloud Storage Bucket

Click *Check my progress* to verify whether the bucket is created.



Create a Cloud Storage Bucket

Check my progress

Train the Model on Vertex AI

Click *Check my progress* to verify the objective.



Train the Model on Vertex AI

Check my progress

Deploy the Model

Click *Check my progress* to verify the objective.



Deploy the Model

Check my progress

Congratulations!

This concluded the self-paced lab, Classify Images with TensorFlow Convolutional Neural Networks. You launched the convolutions notebook and explored convolutions and pooling.

Next steps / learn more

- Learn more about TensorFlow convolutional neural networks.
- Read about Creating art through a Convolutional Neural Network.
- Wonder how AI is being used, see How Moorfields is using AutoML to enable clinicians to develop machine learning solutions or How Google Cloud’s AI has boosted Netmarble’s team collaboration, game development and consumer reach.

Google Cloud training and certification

...helps you make the most of Google Cloud technologies. Our classes include technical skills and best practices to help you get up to speed quickly and continue your learning journey. We offer fundamental to advanced level training, with on-demand, live, and virtual options to suit your busy schedule. Certifications help you validate and prove your skill and expertise in Google Cloud technologies.

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